



Sichuan University -
Pittsburgh Institute

ECE0202 Embedded Processors & Interfacing + Lab

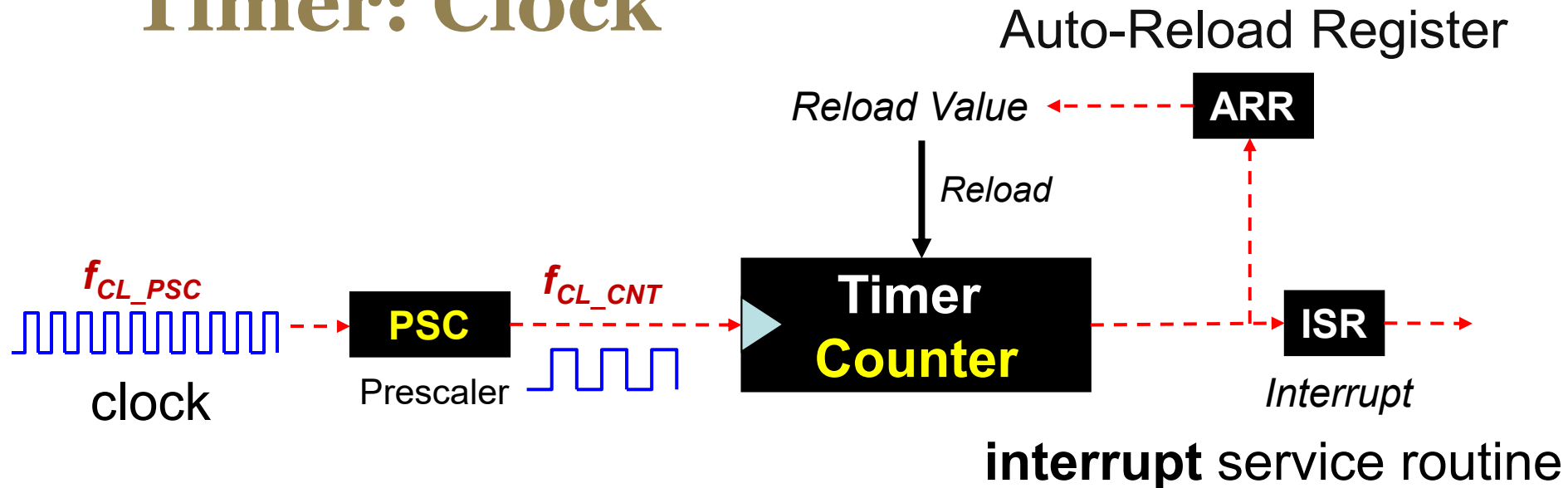
Sichuan University-Pittsburgh Institute



Timer

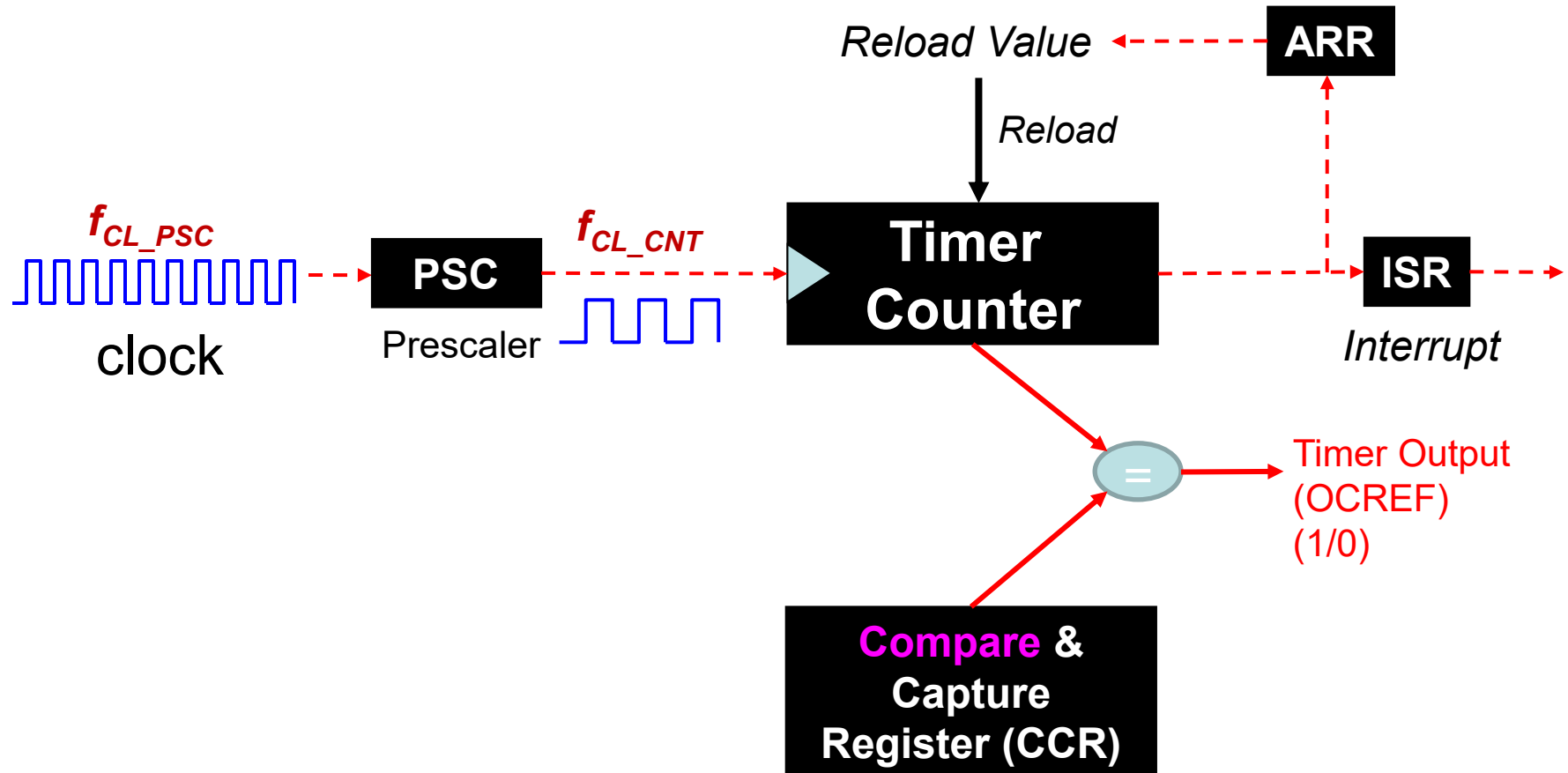
- Free-run counter (independent of processor)
- Functions
 - Input capture
 - Output compare
 - Pulse-width modulation (PWM) generation
 - One-pulse mode output

Timer: Clock

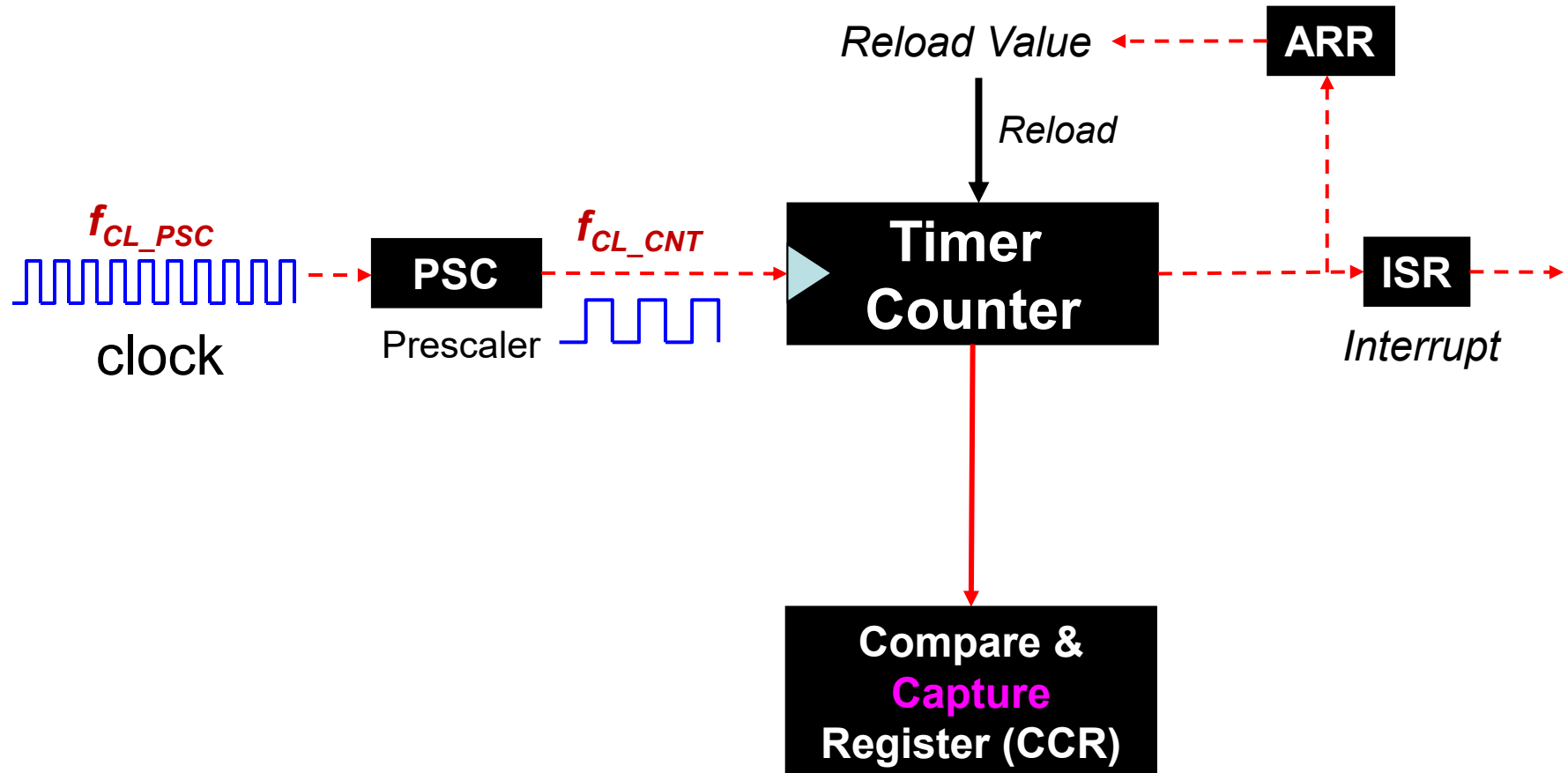


$$f_{CK_CNT} = \frac{f_{CL_PSC}}{PSC + 1}$$

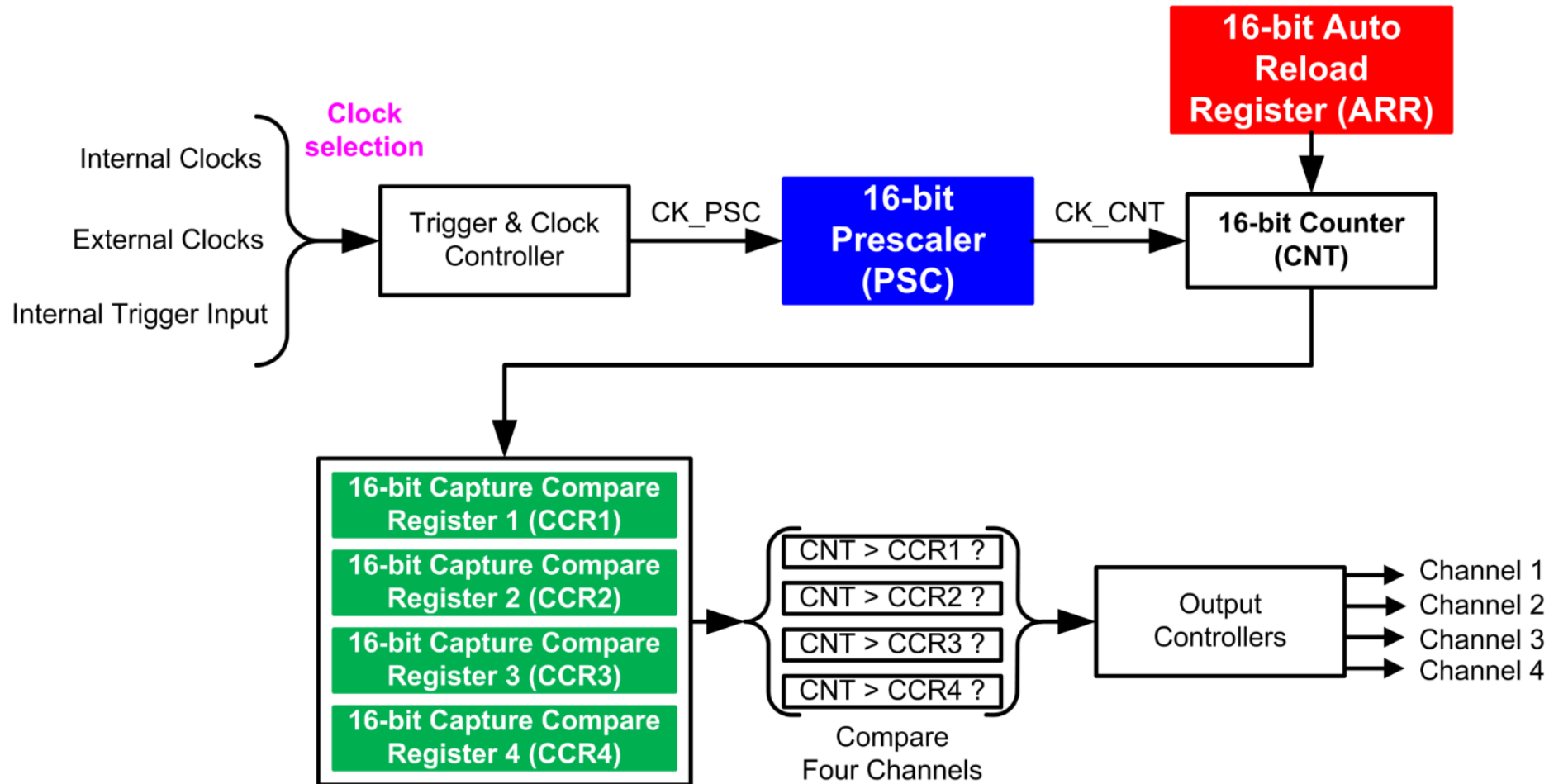
Timer: Output



Timer: Input Capture



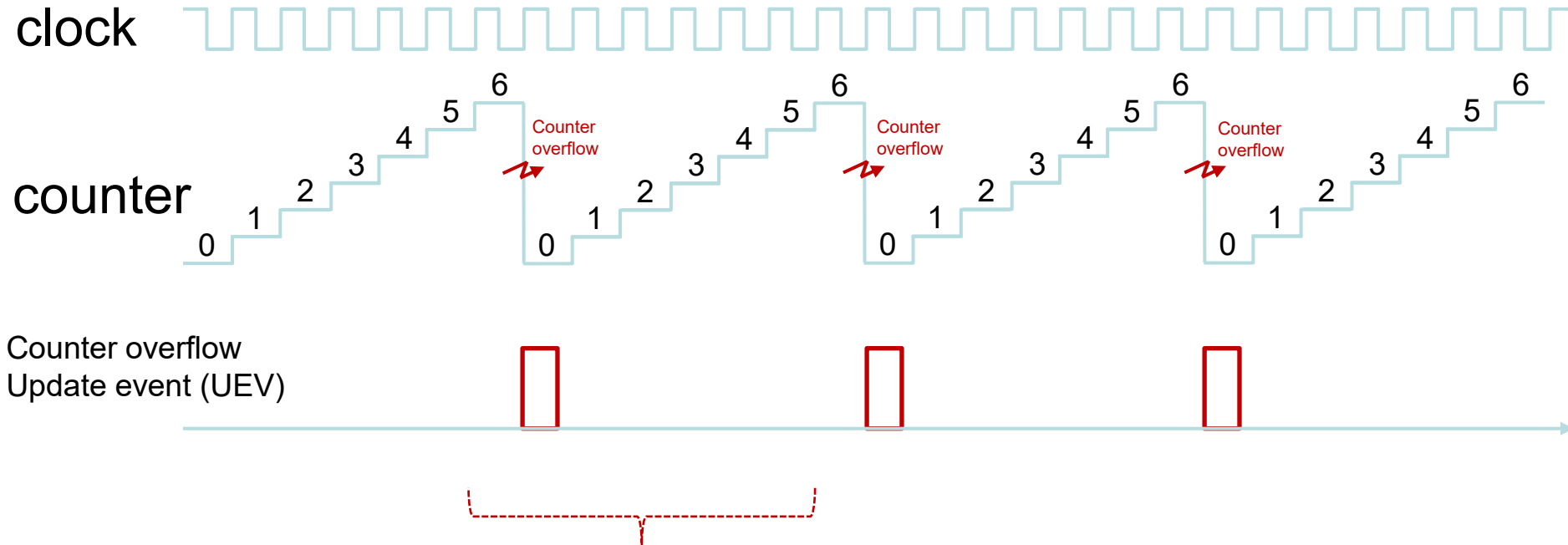
Multi-Channel Outputs



The **duty cycle** of a PWM signal is the relative amount of time the signal will be on and is expressed as a percentage.

Edge-aligned Mode (Up-counting)

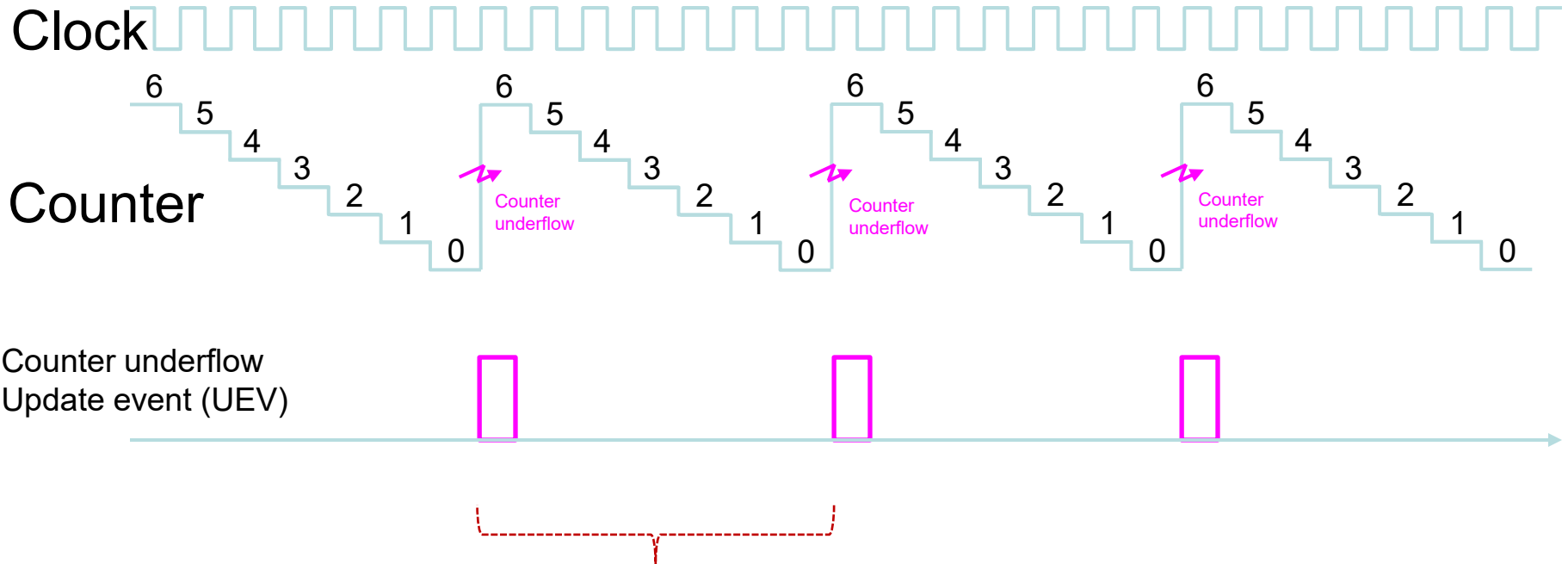
ARR = 6, RCR = 0 repetition-count register



$$\begin{aligned} \text{Period} &= (1 + \text{ARR}) * \text{Clock Period} \\ &= 7 * \text{Clock Period} \end{aligned}$$

Edge-aligned Mode (down-counting)

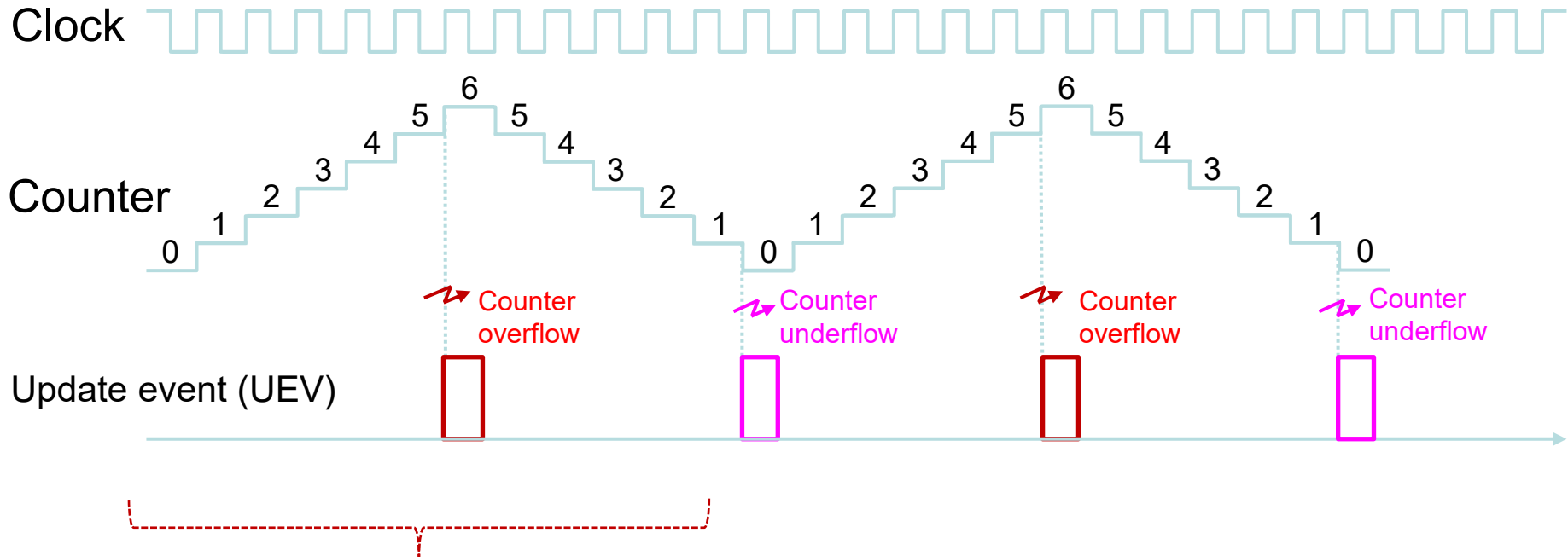
ARR = 6, RCR = 0 repetition-count register



$$\begin{aligned} \text{Period} &= (1 + \text{ARR}) * \text{Clock Period} \\ &= 7 * \text{Clock Period} \end{aligned}$$

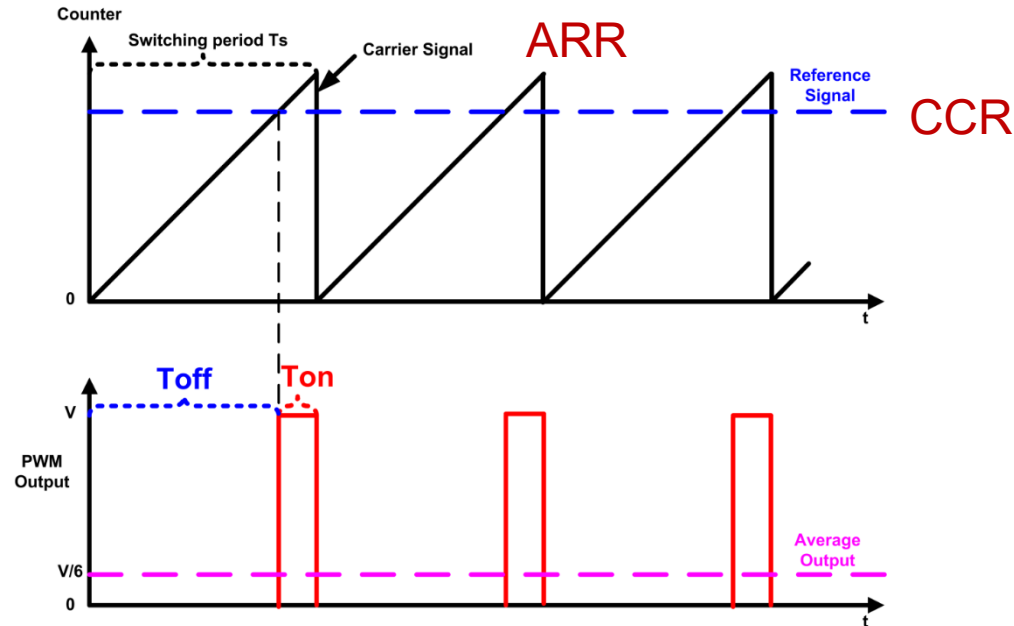
Center-aligned Mode

ARR = 6, RCR = 0 repetition-count register



$$\begin{aligned} \text{Period} &= 2 * \text{ARR} * \text{Clock Period} \\ &= 12 * \text{Clock Period} \end{aligned}$$

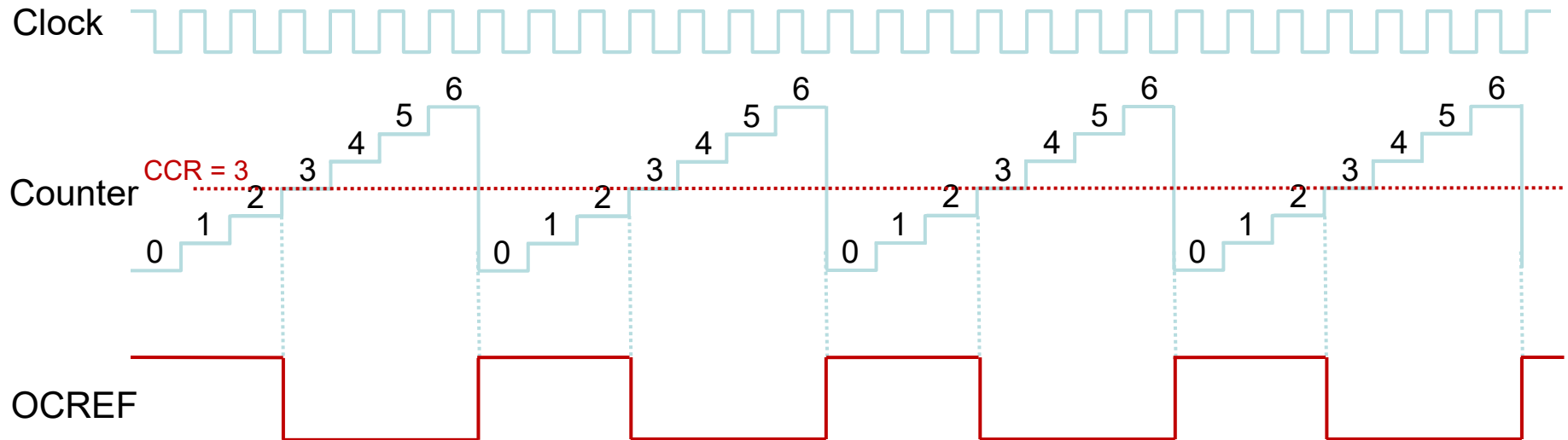
PWM Mode



Mode	Counter < Reference	Counter ≥ Reference
PWM mode 1 (Low True)	Active	Inactive
PWM mode 2 (High True)	Inactive	Active

PWM Mode 1 (Low-True)

Upcounting mode, $ARR = 6$, $CCR = 3$, $RCR = 0$



Period

$$= (1 + ARR) * \text{Clock Period}$$

$$= 7 * \text{Clock Period}$$

Mode 1

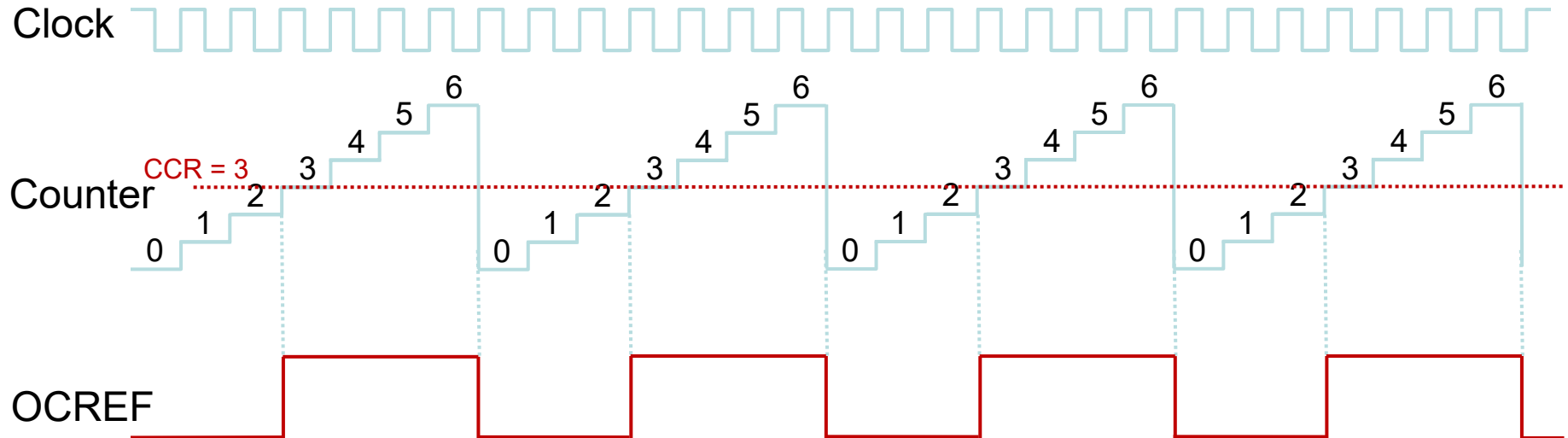
Timer Output =

High if counter < CCR
Low if counter ≥ CCR

$$\text{Duty Cycle} = \frac{CCR}{ARR + 1} = \frac{3}{7}$$

PWM Mode 2 (High-True)

Upcounting mode, $ARR = 6$, $CCR = 3$, $RCR = 0$



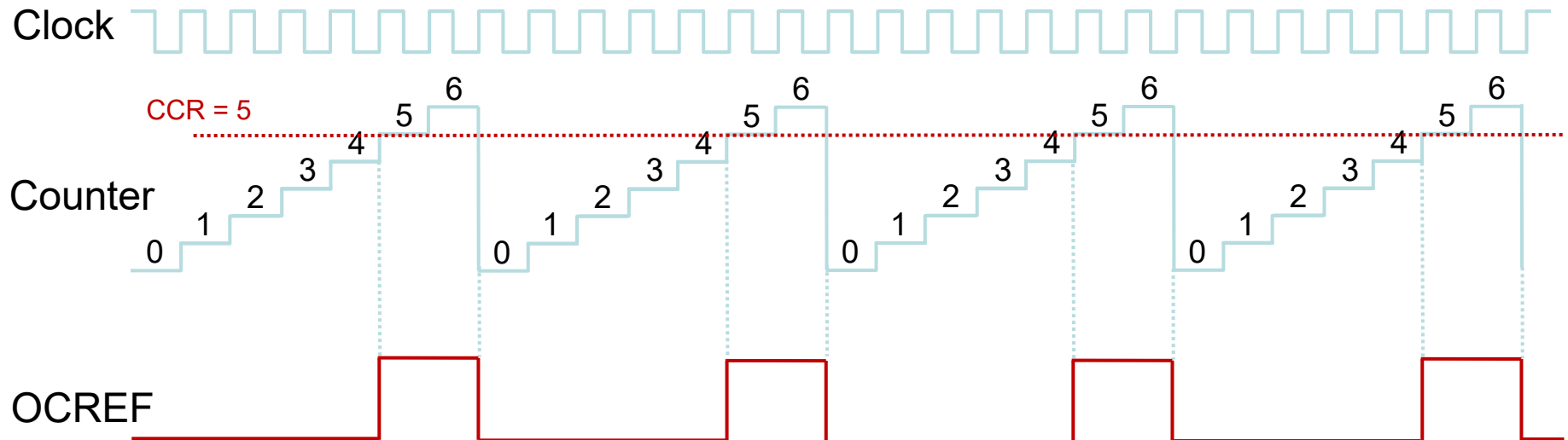
Period
 $= (1 + ARR) * \text{Clock Period}$
 $= 7 * \text{Clock Period}$

Mode 2
 Timer Output = $\begin{cases} \text{Low if counter} < CCR \\ \text{High if counter} \geq CCR \end{cases}$

$$\text{Duty Cycle} = 1 - \frac{CCR}{ARR + 1} = \frac{4}{7}$$

PWM Mode 2 (High-True)

Upcounting mode, $ARR = 6$, $CCR = 5$, $RCR = 0$



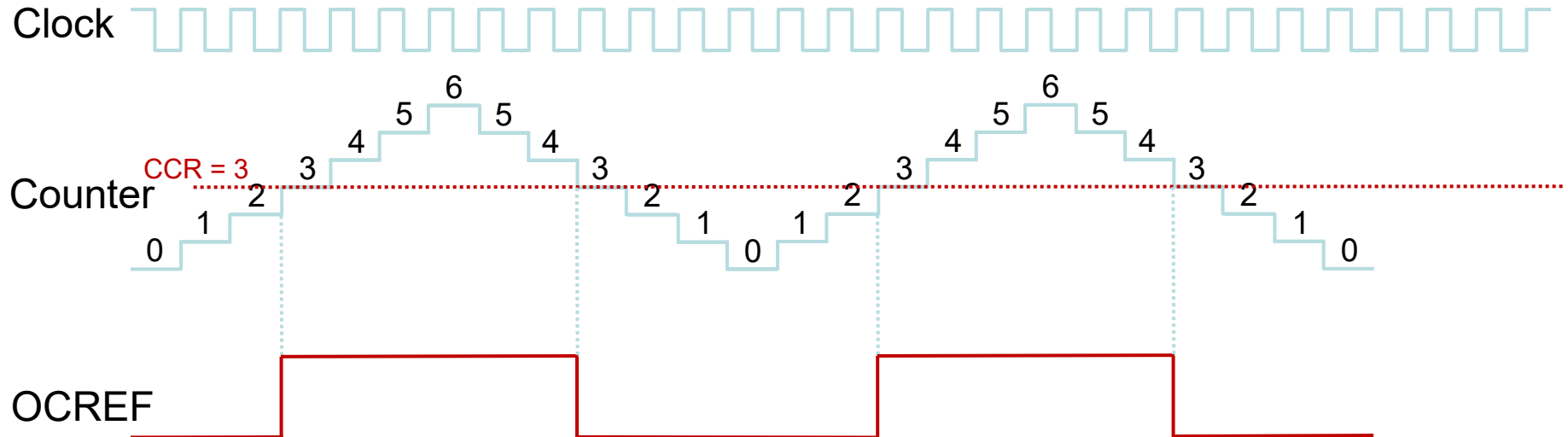
Period
 $= (1 + ARR) * \text{Clock Period}$
 $= 7 * \text{Clock Period}$

Mode 2
 Timer Output = $\begin{cases} \text{Low if counter} < CCR \\ \text{High if counter} \geq CCR \end{cases}$

$$\text{Duty Cycle} = 1 - \frac{CCR}{ARR + 1} = \frac{2}{7}$$

PWM Mode 2 (High-True)

Center-aligned mode, $ARR = 6$, $CCR = 3$, $RCR = 0$



Period

$$= 2 * ARR * \text{Clock Period}$$

$$= 12 * \text{Clock Period}$$

Mode 2

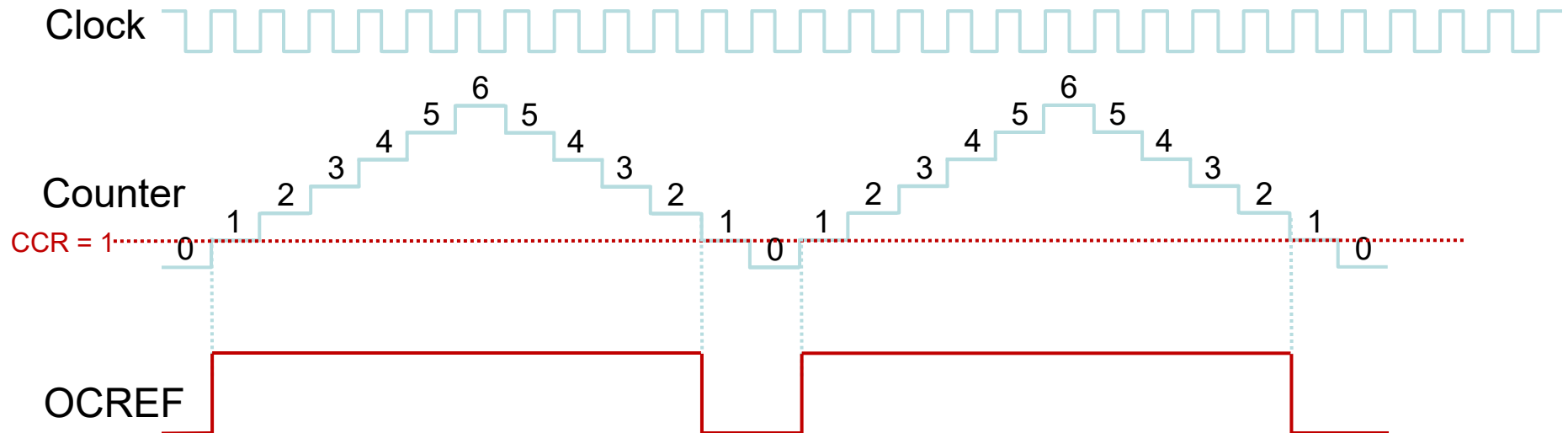
Timer Output =

Low if counter < CCR
High if counter $\geq$ CCR

$$\text{Duty Cycle} = 1 - \frac{CCR}{ARR} = \frac{1}{2}$$

PWM Mode 2 (High-True)

Center-aligned mode, $ARR = 6$, $CCR = 1$, $RCR = 0$



Period
 $= 2 * ARR * \text{Clock Period}$
 $= 12 * \text{Clock Period}$

Mode 2
 Timer Output = $\begin{cases} \text{Low if counter} < \text{CCR} \\ \text{High if counter} \geq \text{CCR} \end{cases}$

$$\text{Duty Cycle} = 1 - \frac{CCR}{ARR} = \frac{5}{6}$$

Using a system clock with frequency 16 MHz, to create the following PWM signal.

Upcounting
Mode 1

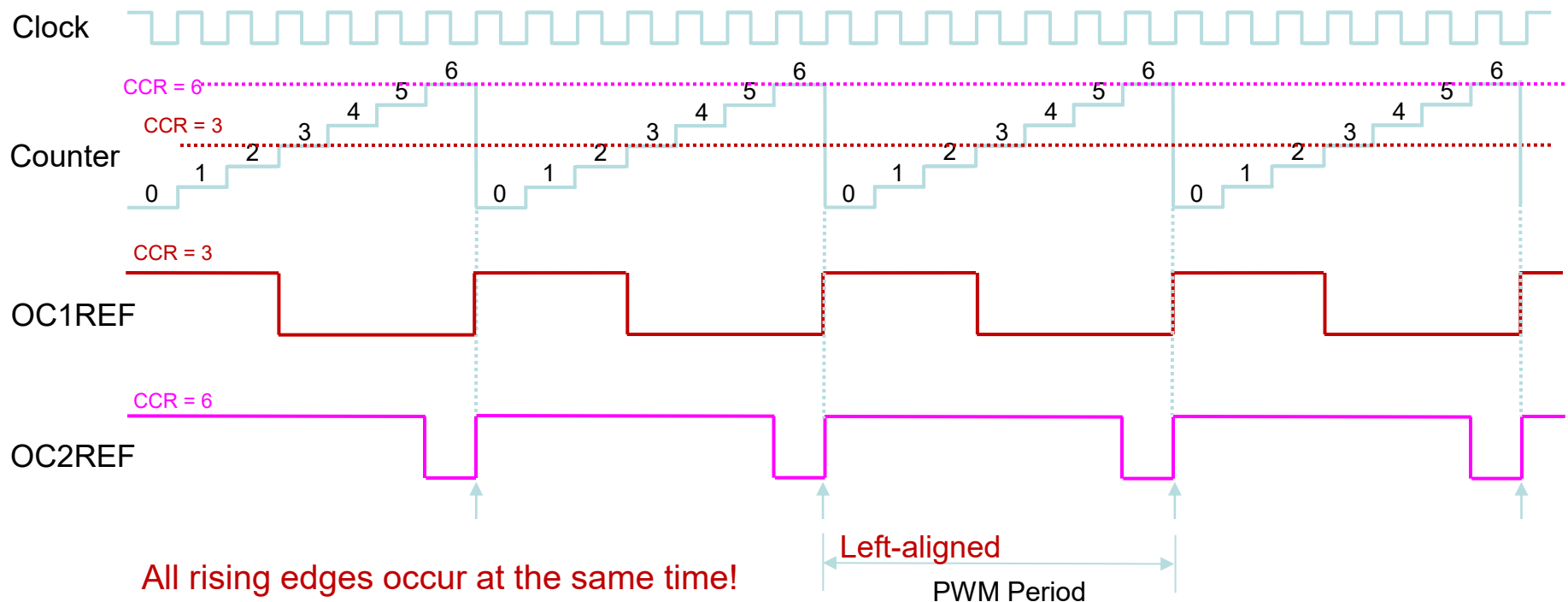
$PSC = 3, ARR = 6, CCR = 3,$

PWM Period = ?

Duty Cycle = ?

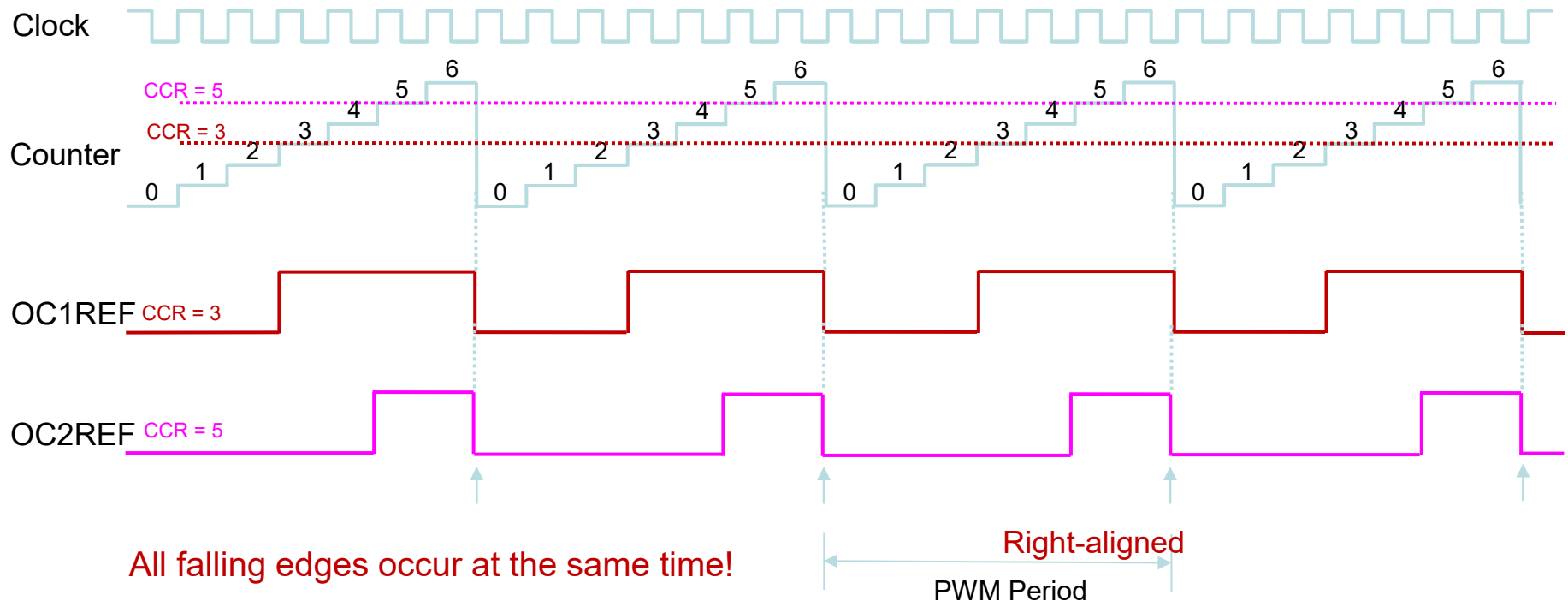
PWM Mode 1 : Left Edge-aligned

Upcounting mode, ARR = 6, CCR = 3, RCR = 0



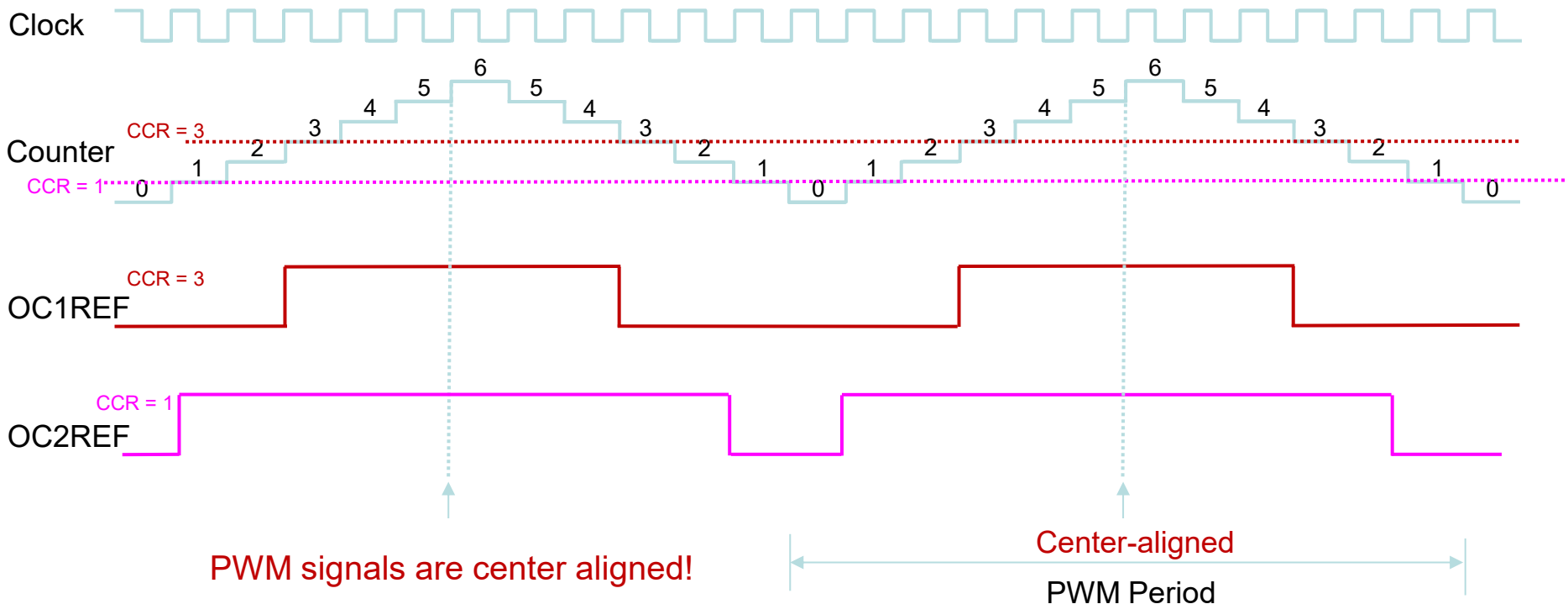
PWM Mode 2: Right Edge-aligned

Upcounting mode, ARR = 6, CCR = 3, RCR = 0



PWM Mode 2: Center Aligned

Center-aligned mode, ARR = 6, CCR = 3, RCR = 0

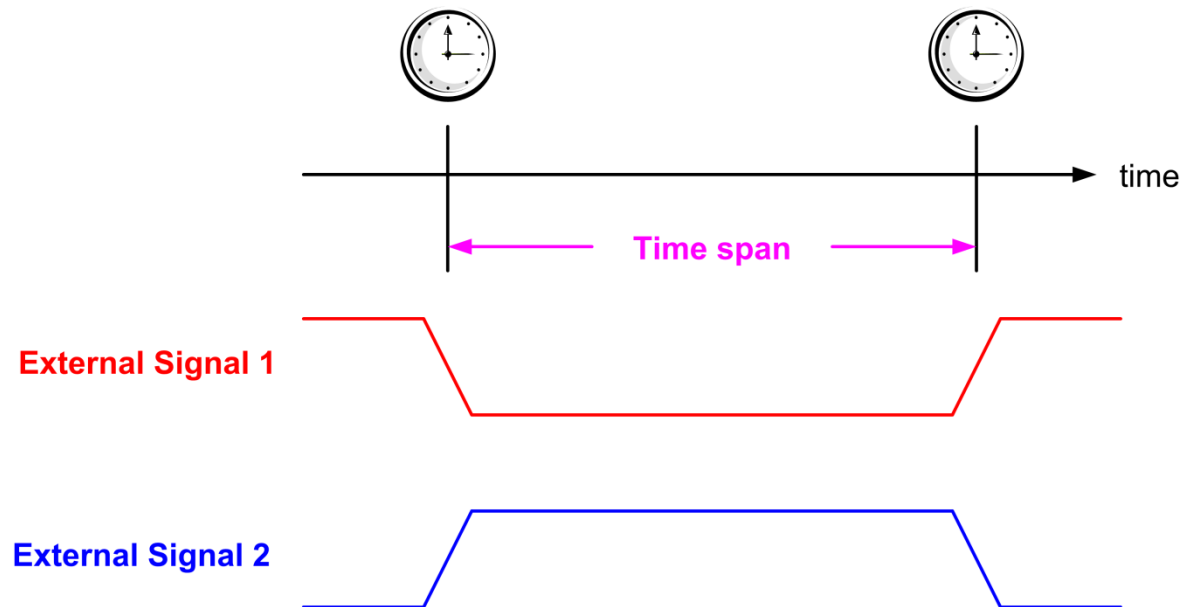


PWM signals are center aligned!

Timer Output =
High if counter $\geq$ CCR
Low if counter $<$ CCR

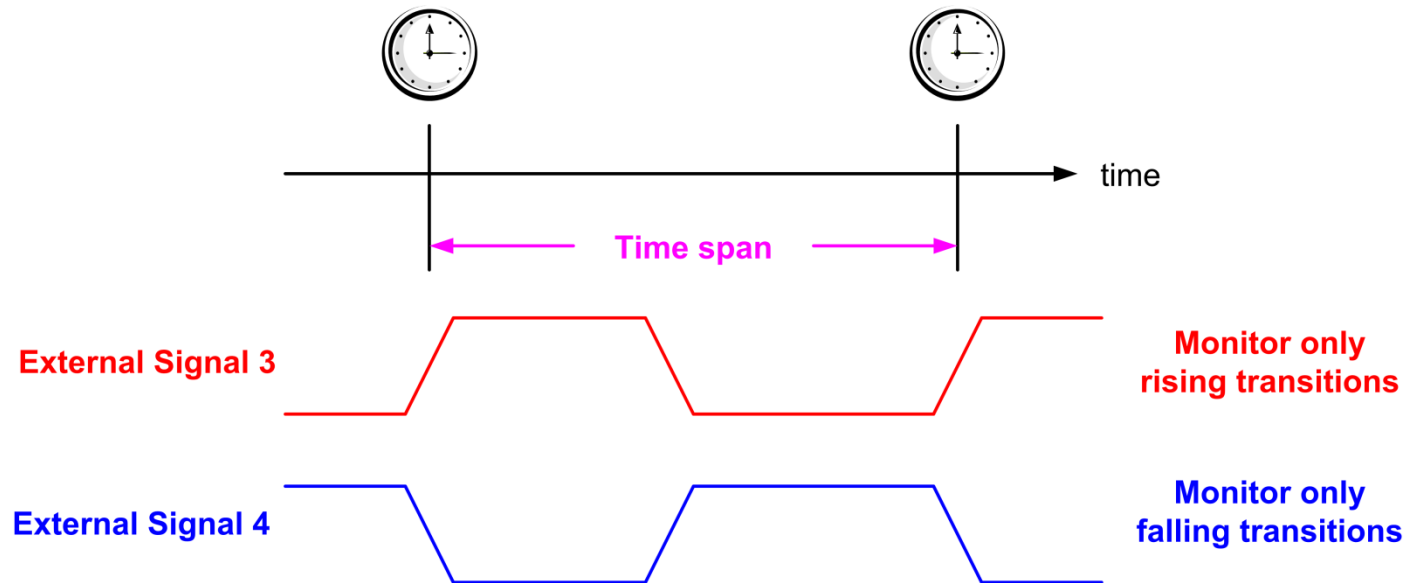
Input Capture

- Monitor both rising and falling edge

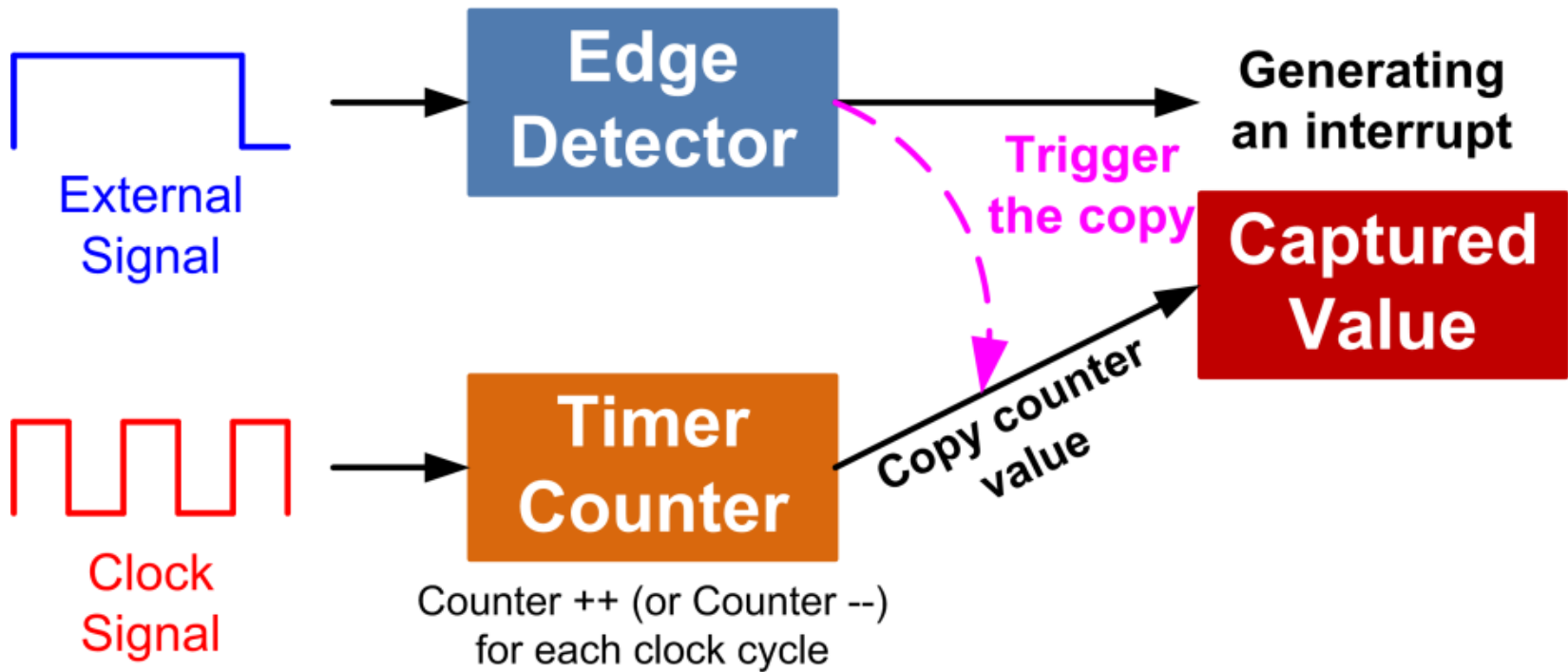


Input Capture

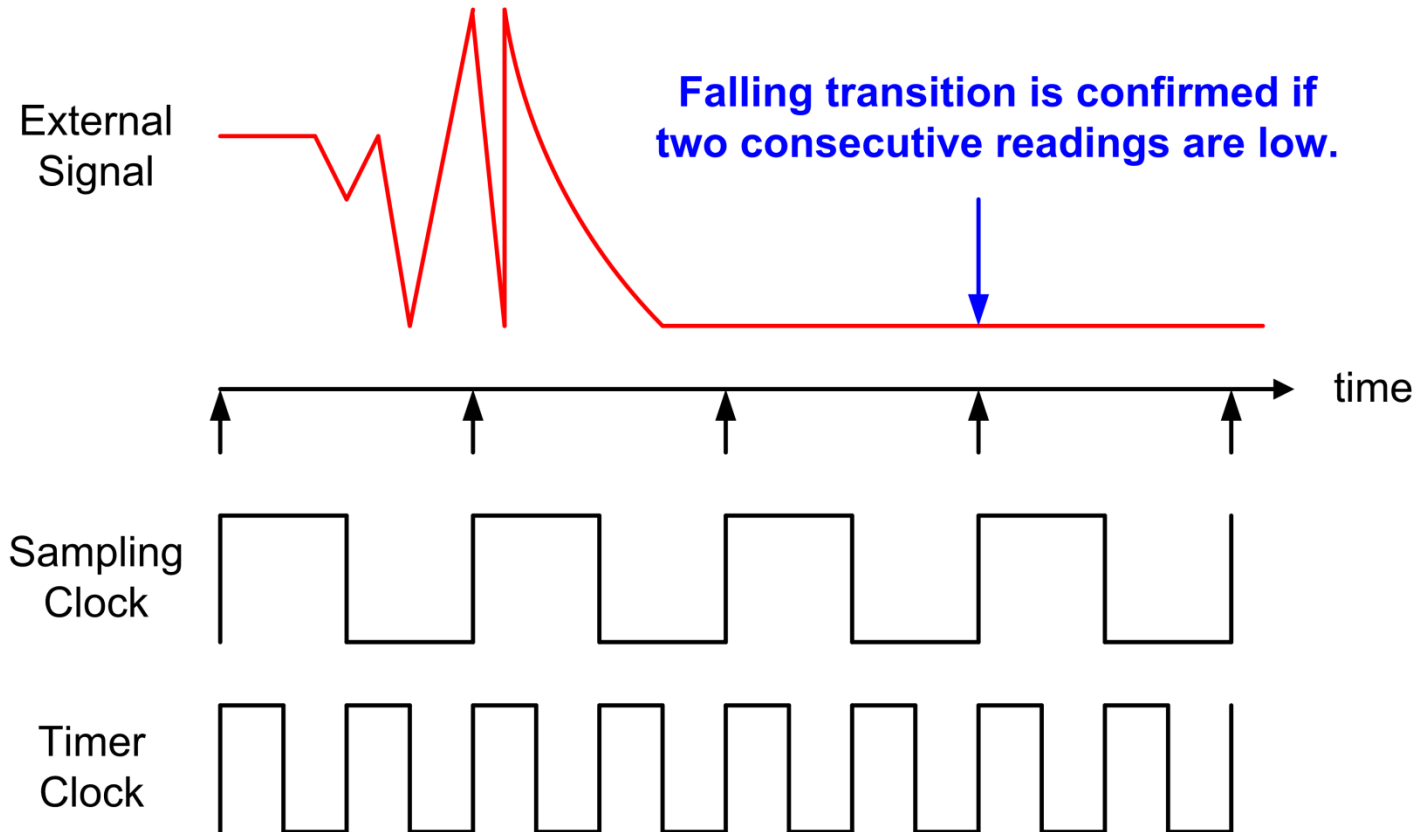
- Monitor only rising edges or only falling edge



Input Capture

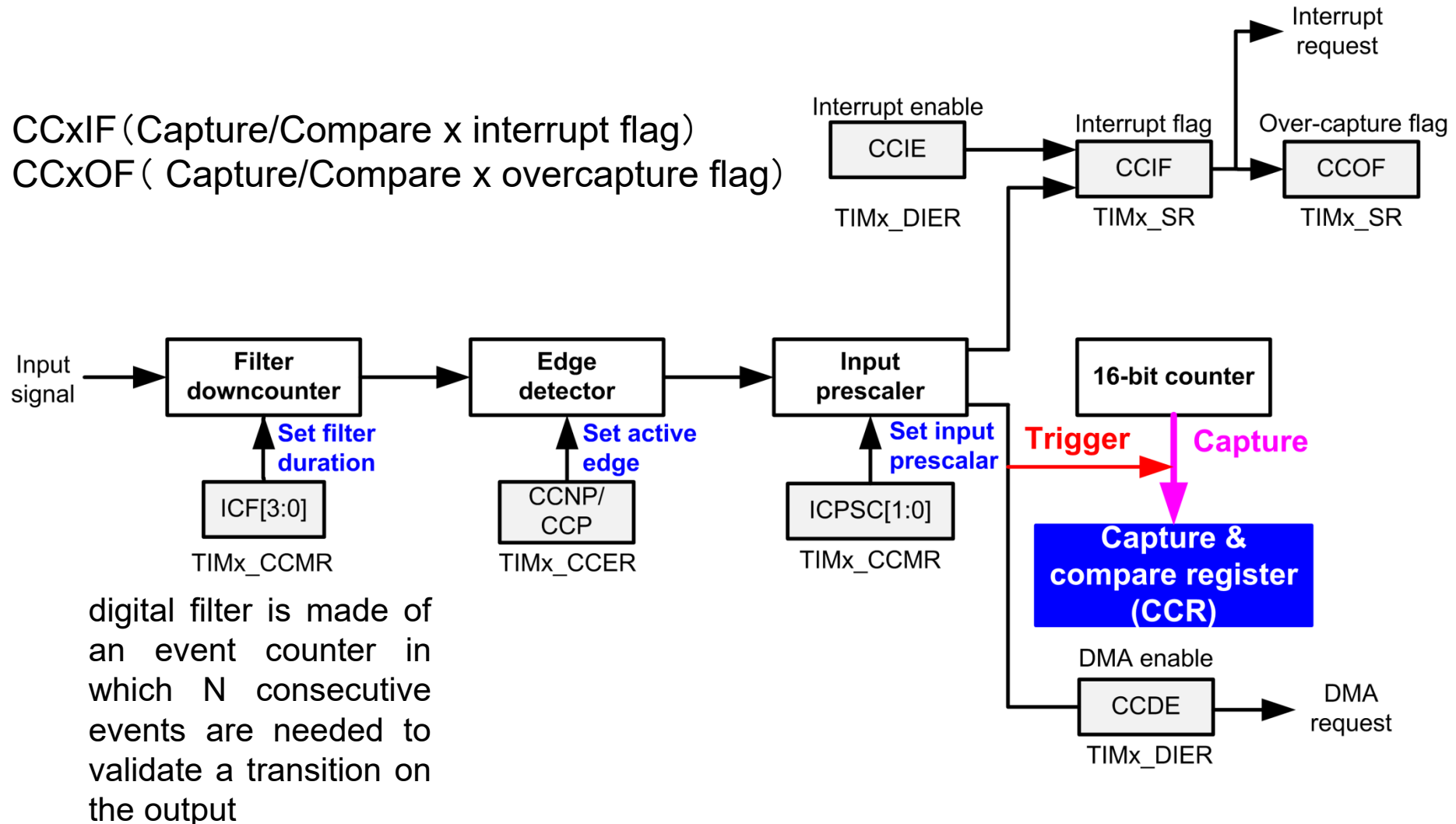


Input Filtering

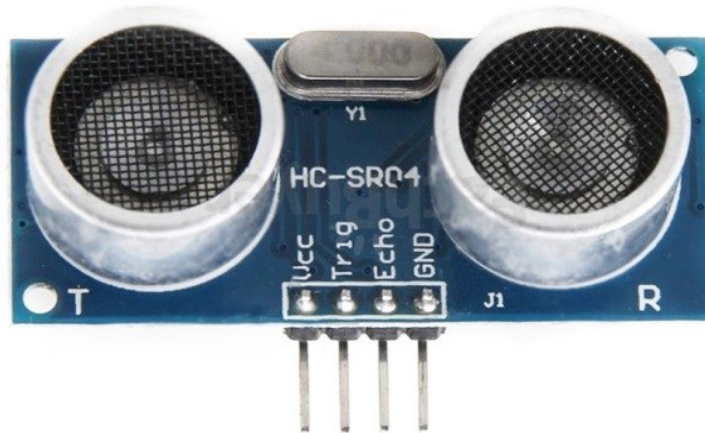


Input Capture Diagram

CCxIF (Capture/Compare x interrupt flag)
CCxOF (Capture/Compare x overcapture flag)



Ultrasonic Distance Sensor



$$\text{Distance} = \frac{\text{Round Trip Time} \times \text{Speed of Sound}}{2}$$

$$= \frac{\text{Round Trip Time}(\mu\text{s}) \times 10^{-6} \times 340\text{m/s}}{2}$$

$$= \frac{\text{Round Trip Time}(\mu\text{s})}{58}$$

Ultrasonic Distance Sensor

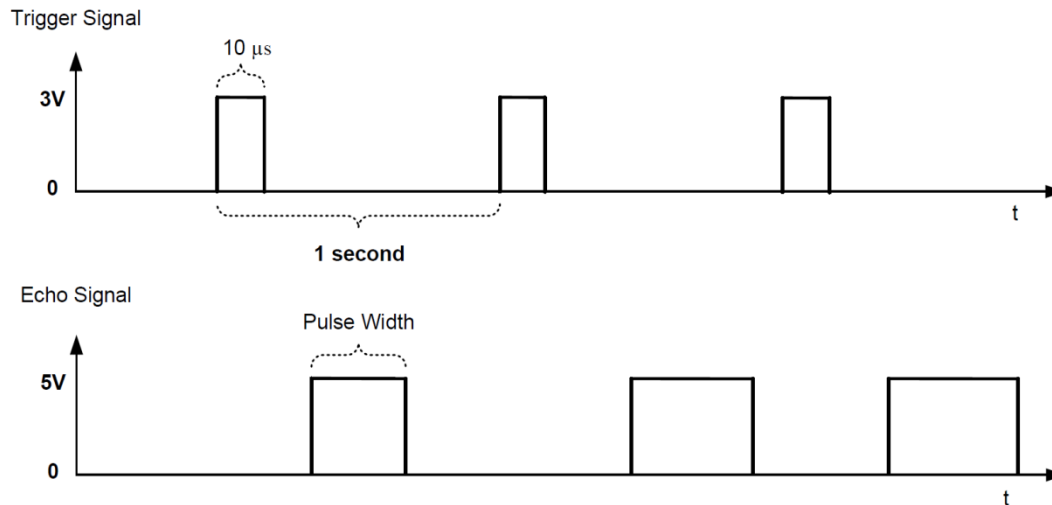


The echo pulse width corresponds to round-trip time.

$$Distance (cm) = \frac{Pulse Width (\mu s)}{58}$$

or

$$Distance (inch) = \frac{Pulse Width (\mu s)}{148}$$



If pulse width is 38ms,
no obstacle is detected.

Ultrasonic Distance Sensor

